

Group 18 Tek4dev 5.1

SME CAPSTONE PROJECT

AI-POWERED BUSINESS INTELLIGENCE DASHBOARD

SMART ANALYTICS FOR SMES IN NIGERIA



FINAL CAPSTONE PROJECT

TEAM: GROUP 18

LEARNING TRACKS: DATA ANALYSIS
& TECHNICAL PROJECT MANAGEMENT



DATA ANALYSIS



BI DASHBOARD



SME GROWTH



BUSINESS INSIGHTS

INNER SPACE INTERIOR DESIGN COMPANY

TABLE OF CONTENT

1. Cover Page & Capstone Introduction

Project Context: Group 18 Tek4dev 5.1 SME Capstone Project.

Strategic Scope: AI-Powered Business Intelligence and Smart Analytics for Nigerian SMEs.

2. Executive Performance Overview

High level financial KPIs (Revenue, Profit, Margins, Total Projects).

Revenue trends over time, Property Type distribution, and Decomposition Tree analysis.

3. Product & Services Analysis

Service category revenue contributions (Painting, Renovation, Decoration, etc.).

Residential vs. Commercial sector analysis.

Cost vs. Revenue distribution matrix and profit margin scatter profiles.

4. Customer & Regional Insights

Geographic footprint across Nigerian states.

Lead generation tracking by referral source performance.

Client payment status matrices (Outstanding, Paid, Part Payment).

Operational pipeline health (Completed, Ongoing, Delayed, and Cancelled metrics) with interactive NLP Q&A integration.

- Key Insights

- Recommendations

Executive Performance Dashboard

Total Revenue

₦2.48bn

Total Profit

₦716.85M

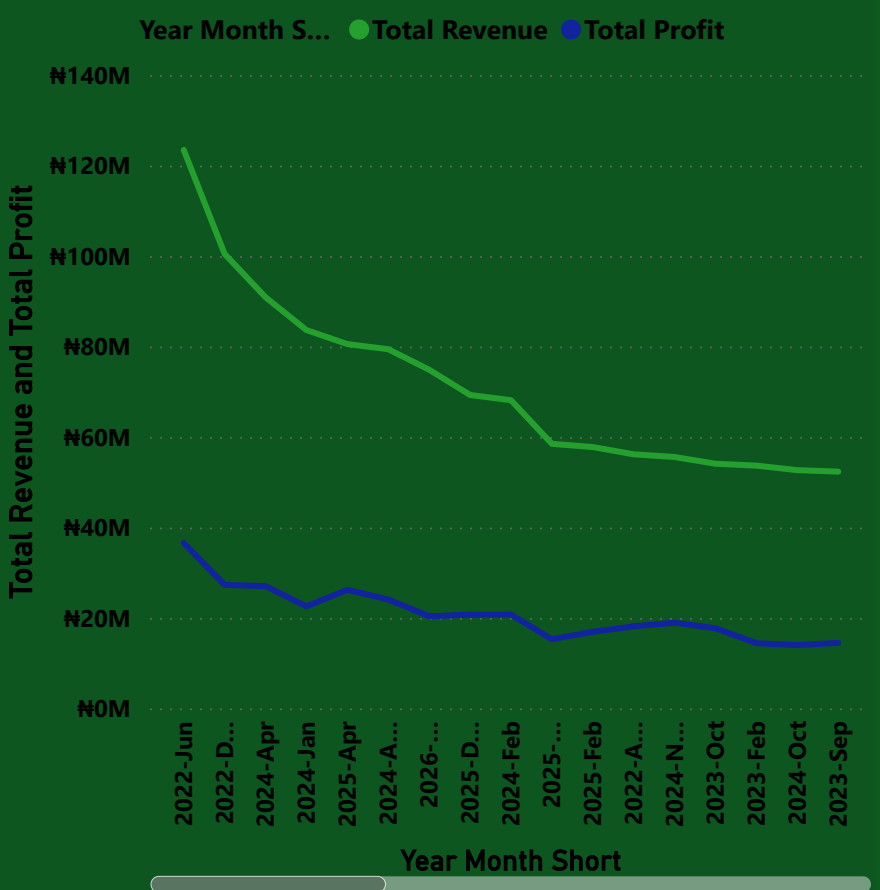
Overall Profit Margin

28.95%

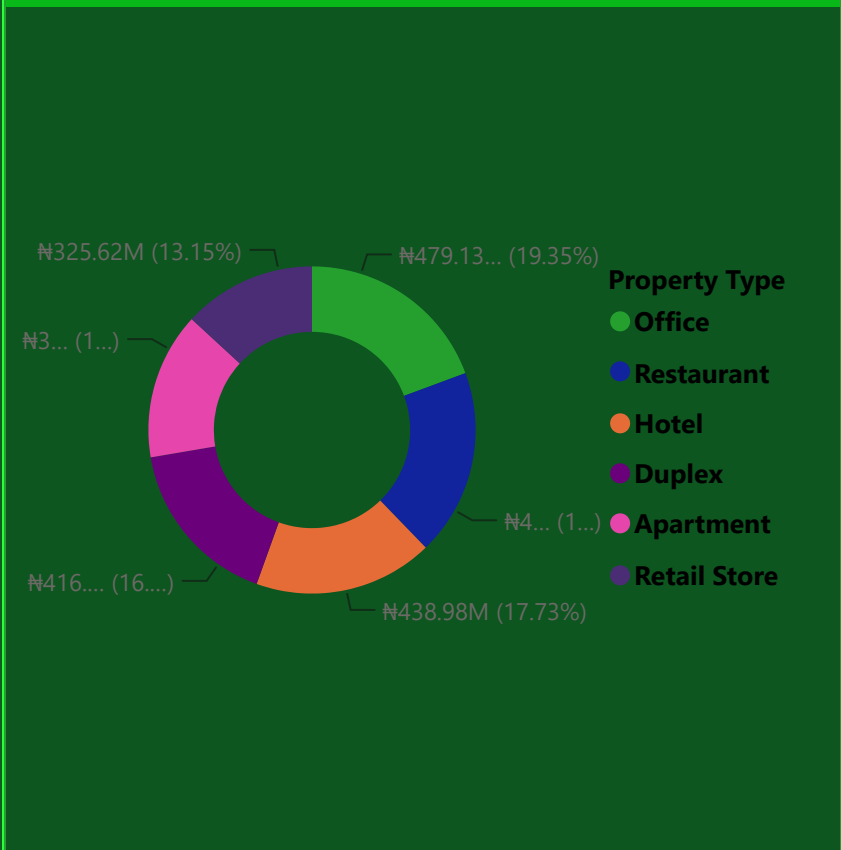
Volume of Projects Managed

299

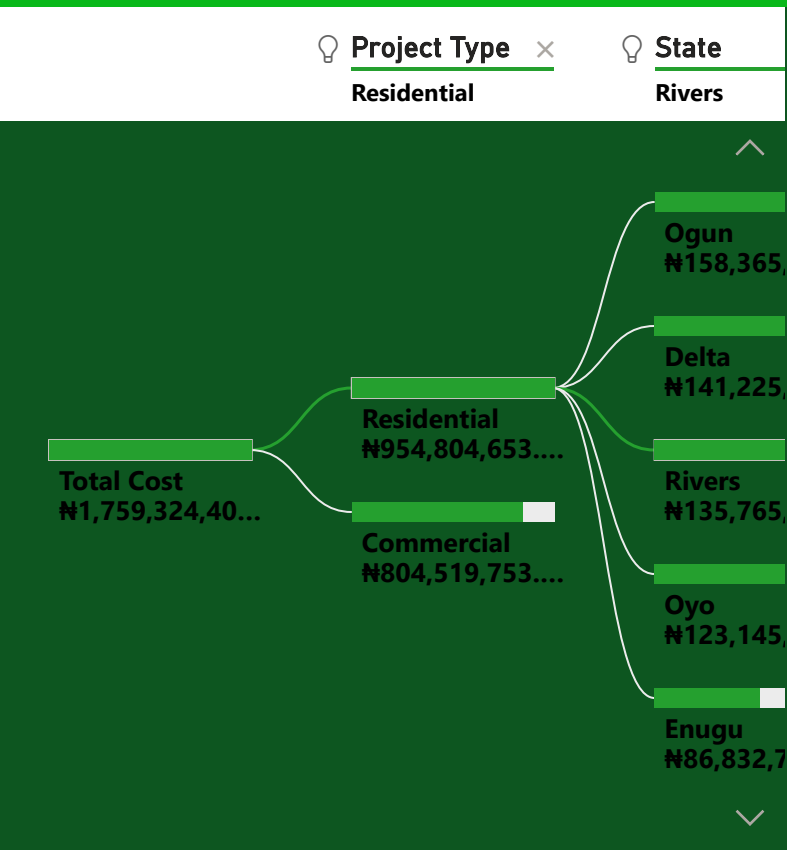
Monthly Revenue & Profit Growth Trend



Revenue Ingestion: Commercial & Residential

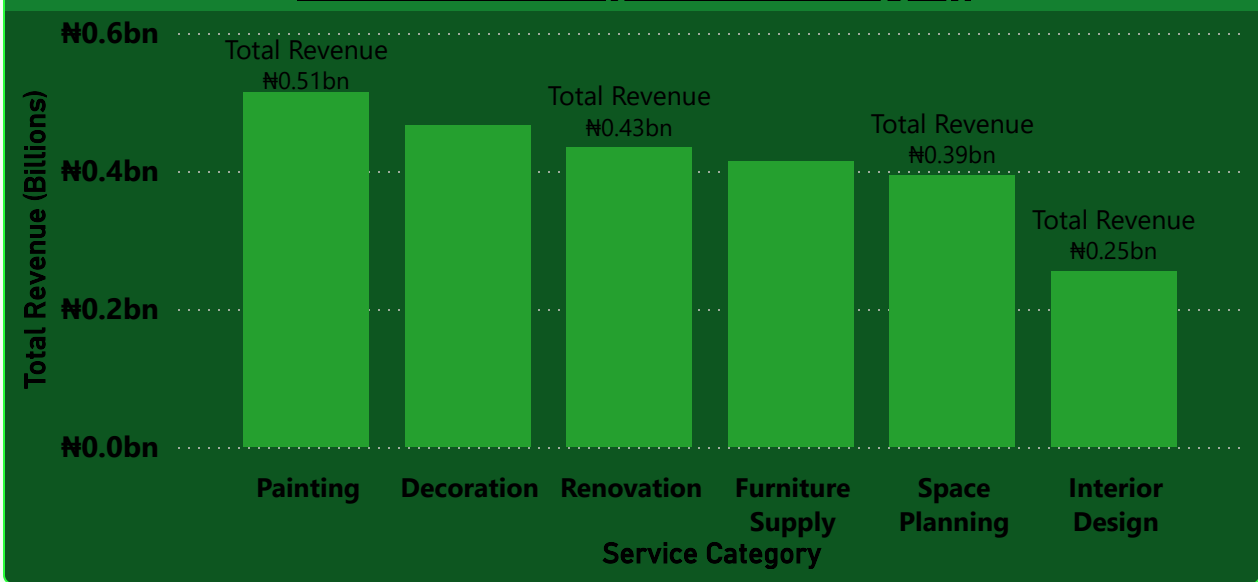


Collection Efficiency and Payment Status Pipeline

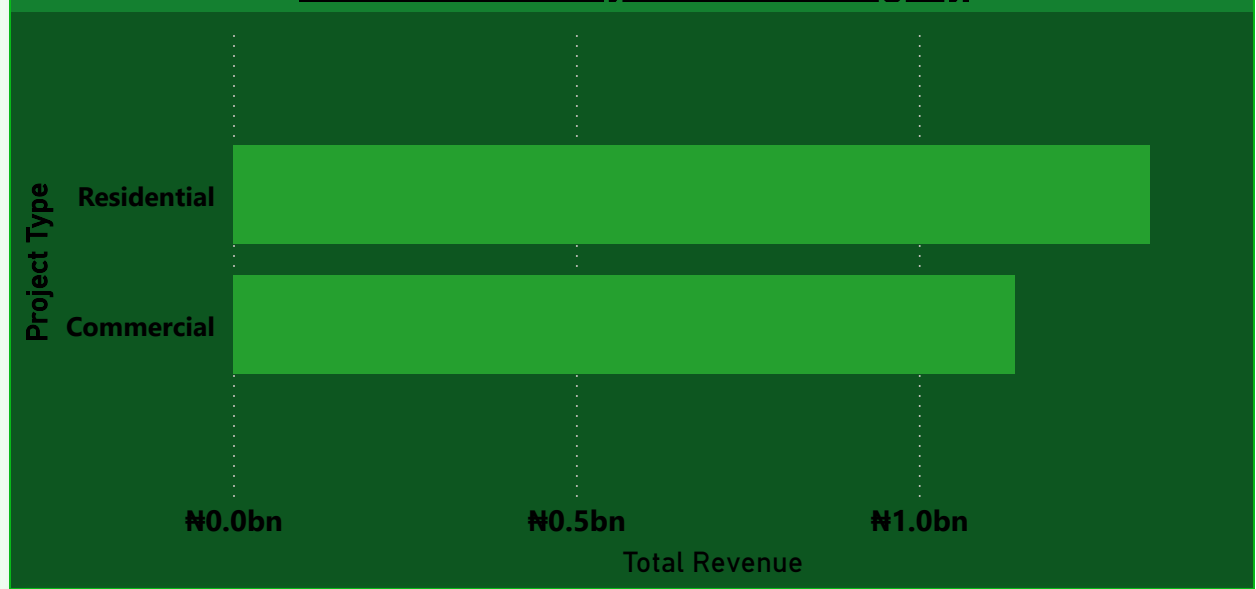


Product & Services Analysis

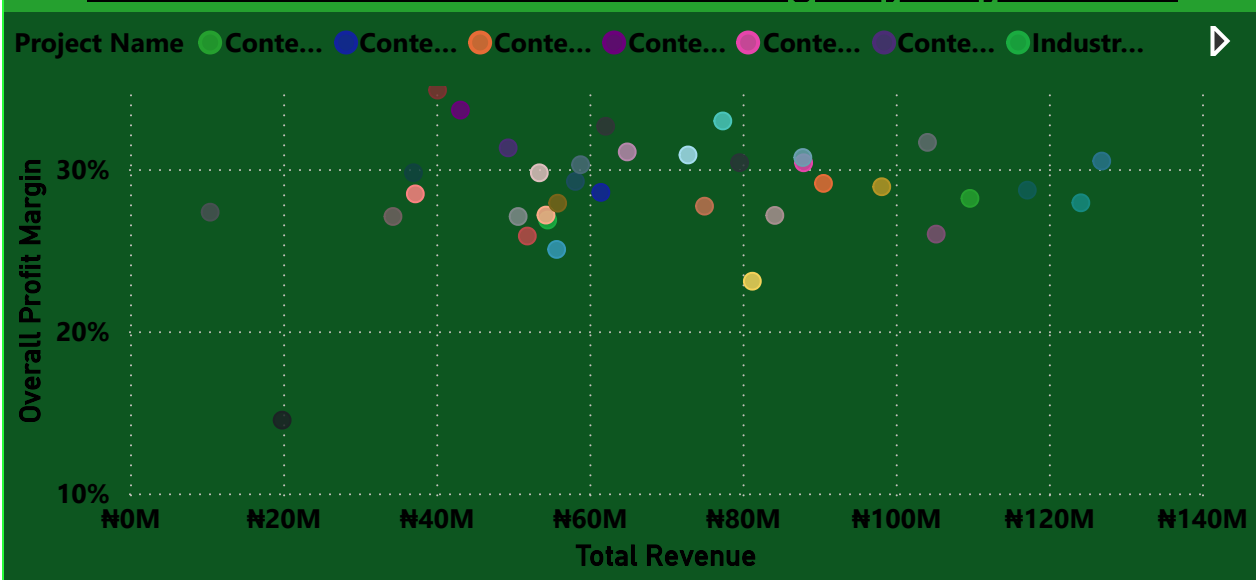
Financial Yield by Service Category



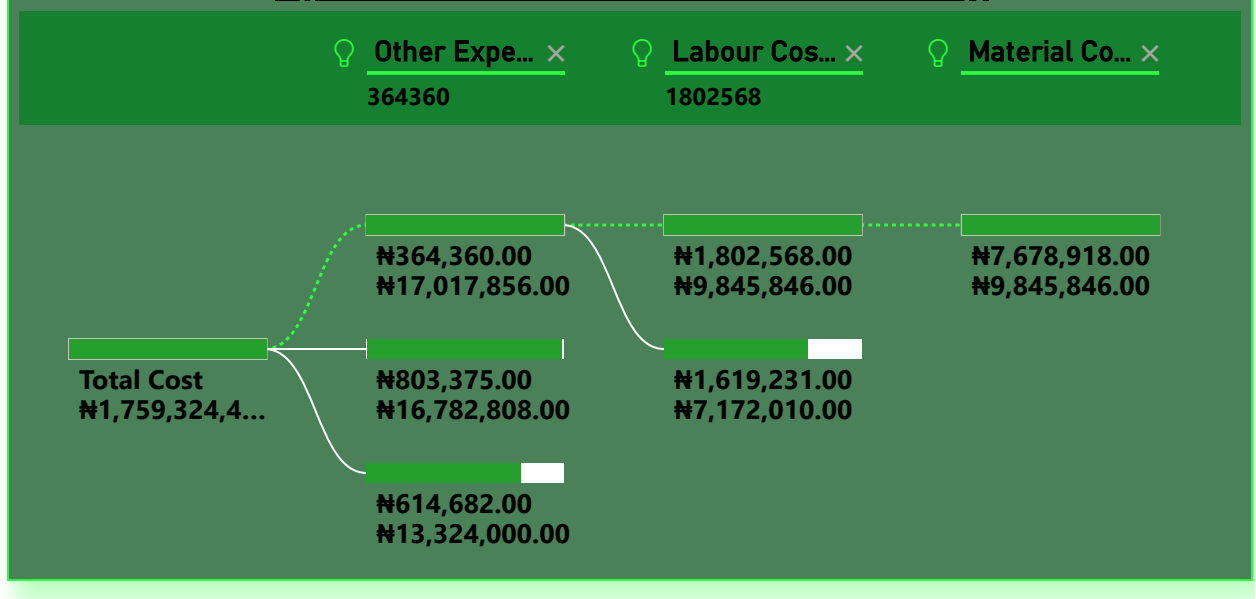
Revenue Yield by Service Category



Total Revenue and Overall Profit Margin by Project Name



Operation Cost Breakdown Hierarchy

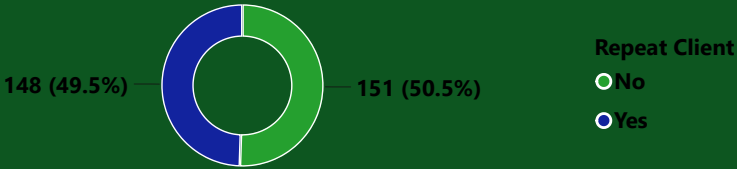


Customer and Regional Insights

Geographic ingestion Footprint by Nigerian State



Distribution Matrix of Customer Satisfaction Ratings

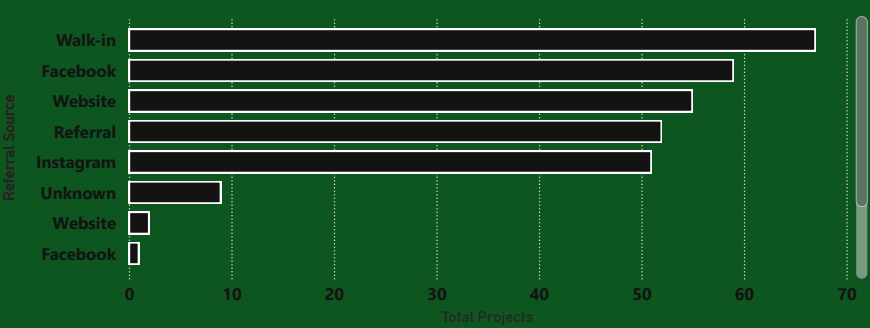


Customer Matrix: Account, Retention and Satisfaction Metrix

Client Type	Outstanding	Paid	Part Payment	Total
Individual	N353,170,453.00	N398,208,832.00	N332,934,863.00	N1,084,314,148.00
Corporate	N489,691,331.00	N427,145,307.00	N475,023,409.00	N1,391,860,047.00
Total	N842,861,784.00	N825,354,139.00	N807,958,272.00	N2,476,174,195.00

Lead Generation: Project Volumes by Referral Source

Total Projects by Referral Source



Project Pipeline and Revenue Breakdown

Total Revenue	Completed Projects	Ongoing Projects	Delayed Projects	Cancelled Projects
N2,476,174,195.00	137	6	75	81

Questionnaire

Q&A will be retiring in December 2026.

Remove Q&A

Top client name

Showing results for *Top client name of projects by total estimated budget(?)*

Patricia Brown

\$199,076,166

Sum of Estimated Budaet(?)

Key Findings

1. Financial Baseline & Profitability

The business has generated a healthy ₦2.48 Billion in Total Revenue with a ₦716.85 Million Total Profit.

This places the overall company profit margin at a stable 28.95% across its 299 total projects.

2. Product and Sector Dominance

Residential projects outperform Commercial ones, securing a higher share of total revenue.

Painting is the top-earning service category (bringing in roughly ₦0.51 Billion), closely followed by Decoration and Renovation, while core Interior Design services represent the lowest category revenue stream at ₦0.25 Billion.

Within property types, Office spaces and Retail stores anchor a massive chunk of the pie chart, indicating strong corporate contract sizes despite higher overall residential volumes.

3. Pipeline Bottlenecks & Operational Risks

Out of 299 projects, 137 are Completed. However, there are significant operational risks: 75 projects are currently Delayed and 81 have been Cancelled.

This means over 27% of signed projects are being cancelled, representing a severe leakage in the pipeline that requires immediate operational review.

4. Cash Flow and Payment Exposures

Your customer matrix reveals an invoicing risk. Out of the ₦2.48 Billion total pipeline value, ₦842.86 Million remains Outstanding.

Corporate clients hold the highest individual category revenue (₦1.39 Billion), but they also hold the largest chunk of unpaid debt (₦489.69 Million Outstanding), creating a potential working capital constraint for an SME.

Data-Driven Recommendations for the SME

1. Implement an Automated Milestones Alert System

The Insight: With 75 projects currently delayed, project timelines are slipping before management can intervene.

The Action: Use the data pipeline to flag projects that have spent more than a specific number of days in the "Ongoing" phase without a status change. Set up proactive triggers (via Power Automate if integrating with their workflow) to alert project managers 5 days before a milestone deadline is breached.

2. Revamp the Corporate Client Onboarding & Retainer Policy

The Insight: Your customer matrix shows that corporate clients hold a massive ₦489.69 Million in outstanding debt. Financial constraints often lead to operational delays or outright project cancellations.

The Action: Tie project kickoff milestones strictly to cash collection. Implement a policy requiring a higher upfront deposit (e.g., 50%) from corporate clients, or halt design phases immediately if a "Part Payment" milestone is overdue by more than 14 days.

3. Audit Low Performing Referral Sources

The Insight: Your "Total Projects by Referral Source" bar chart exposes where leads are coming from. High-volume channels with high cancellation rates indicate poor lead quality.

The Action: Cross reference your Cancelled Projects measure against the Referral Source column in a matrix table. If specific digital channels (like Facebook or Instagram) show a disproportionately high rate of cancellations, recommend shifting the marketing budget toward higher-intent channels like your website or direct client referrals.

1. Optimize Supply Chain and Vendor Management for Key Services

The Insight: Painting, Decoration, and Renovation are your highest revenue generating services. Bottlenecks here heavily impact overall delivery timelines.

The Action: Delays in interior design often stem from material stockouts (paints, furniture items, fixtures) or contractor availability. The business should establish SLA (Service Level Agreement) contracts with reliable local Nigerian vendors to guarantee priority delivery windows for high volume materials.